Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/12/2016
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Modeling of Particle Transport in the Melt and Its Interaction with the Solid-Liquid Interface		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	derby@umn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	612-625-8881
Organization Name:	University of Minnesota		
PI Address 1:	Department of Chemical Engineering and Materials Science		
PI Address 2:	151 Amundson Hall, 421 Washington Avenue, SE		
PI Web Page:			
City:	Minneapolis	**State:**	MN
Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/01/2010	**End Date:**	11/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
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Flight Assignment:	NOTE: End date is 11/30/2015, per NSSC information (Ed., 6/3/15) NOTE: End date is 8/31/2015, per NSSC information (Ed., 6/3/14) NOTE: End date is 8/31/2014, per NSSC information (Ed., 5/30/2013)		
Key Personnel Changes/Previous PI:	Jeffrey Derby is the PI of this NASA project. Collaborators funded by the DLR ParSiWal project are Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth).		
COI Name (Institution):			
Grant/Contract No.:	NNX10AR70G		
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Task Description:	The primary goal of this research project is to understand the mechanisms that determine the dynamics of particle transport in silicon melts and the pushing or engulfment of these particles at a moving solid-liquid interface, specifically small silicon carbide and silicon nitride particles during the solidification of silicon. This understanding is vital to improving the quality of solar-grade, multi-crystalline silicon grown by directional solidification processes. This work will coordinate with the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR (Deutsches Zentrum für Luft- und Raumfahrt) and involving Jochen Friedrich (Fraunhofer IISB), Arne Croell (University of Freiburg), and Heike Emmerich (University of Bayreuth). Specifically, finite-element, moving-boundary analyses are being developed and applied to assess the dynamics of particle pushing and engulfment. Of particular interest is to ascertain critical velocities for engulfment for a variety of different particle sizes and thermal gradients. We will also study the effects of shear and solidification flows, in the melt near the interface, on engulfment mechanisms and the effect of a grain boundary at the interface that locally modifies its shape and energy. Planned experiments by ParSiWal in the laboratory as well as during the TEXUS mission are aimed to probe the effects of particle engulfment during silicon solidification and will be used to assess theoretical predictions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the dynamics of particle pushing and engulfment will enable process development to minimize their effects on grown material. This will promote lower cost and higher quality of solar-grade, multi-crystalline silicon and will advance progress in commercial solar cell production.
Task Progress:	When casting multicrystalline silicon (mc-Si) ingots for solar cells, the silicon melt usually contains a certain concentration of carbon and nitrogen, which both possess relatively low segregation coefficients in silicon [1]. During directional solidification, these two elements will accumulate to supersaturation levels in the liquid silicon and precipitate to form solid particles of silicon carbide (SiC) and silicon nitride (Si3N4). Depending upon solidification conditions, these foreign particles in the melt can either be pushed or be engulfed by the advancing solidification front. The engulfed particles, which eventually become inclusions in mc-Si ingots, can lead to lower cell efficiency, wafer breakage, and sawing defects. Even worse, when slicing the ingots into wafers, the sawing wire can be broken by these extremely hard inclusions [2], which significantly increases costs. There is a long history and substantial research on the analysis of particle engulfment during solidification, because this topic is also important in other physical processes, such as fabrication of metal-matrix composites, separation processes, cryogenic preservation of biological materials, and frost heaving. Pioneering studies on modeling the particle engulfment processes have been performed by Uhlmann, Chalmers, and Jackson [3] and Chernov, Temkin, and Mel'nikova [4, 5], and particularly relevant research has been more recently put forth by Stefanescu et al. [6], Rempel and Worster [7, 8], Park et al. [9], and Garvin et al. [10, 11]. Only a few researchers have specifically focused on the solidification of silicon, due to the difficulties of high temperatures (greater than 1687K) and small particle sizes (typically ranging between 1–400μm). Among those limited studies on the silicon system, Søiland et al. [12], Trempa et al. [13], and Reimann et al. [14] investigated the occurrence of SiC and Si3N4 inclusions by analyzing ingot samples from direc- tional solidification. Recently, Jauss et al. [15] and Azizi et al. [16] have aimed to quantitatively determine engulfment conditions via experiments using a silicon floating zone system pre-seeded with a distribution of SiC particles and grown under microgravity and terrestrial conditions. These studies of inclusions during silicon crystal growth have pointed out that classical theoretical models of engulfment do a poor job of explaining experimental observations. Therefore, we desire to develop new approaches to model particle engulfment and to further our understanding of this complicated phenomenon in this system. We have performed both steady-state and transient simulations for particle engulfment during directional solidification using finite-element models that rigorously couple heat transfer, fluid flow, and interfacial phenomena. Steady-state analyses show how an unambiguous critical velocity can be determined from solutions for minimum gap thickness versus growth rate. Furthermore, bifurcation theory identifies the critical velocity as a limit point in the steady solution curve, separating branches that correspond to stable and unstable states. We also employ transient computations to demonstrate that, when a steadily solidifying interface approaches a resting particle, the dynamic particle pushing-engulfing transition is consistent with the steady-state critical velocity, as observed in many experiments. Of particular interest and relevance for this research are experiments on the engulfment of SiC particles by solidifying silicon in ground and microgravity flight experiments conducted via the ParSiWal (Partikeleinfang bei der Siliziumkristallisation im Weltall) project funded by the German DLR (Deutsches Zentrum für Luft- und Raumfahrt) and involving Jochen Friedrich (Fraunhofer IISB) and Arne Croell (University of Freiburg). In comparing these data with theoretical predictions, we find that the prediction of critical velocities, based on classical engulfment theory [6], poorly describes the experimental data. Our model better describes the engulfment-pushing transition for smaller particles, and there are several other enlightening observations associated with our model predictions. First, the shapes of the predicted engulfment curves are quite different. In a log-log plot showing engulfment states as functions of critical velocity, Vc, and particle diameter, D, the classical Søiland model predicts a straight line with a slope of -1, indicating that Vc is inversely related to D-1. This inverse relationship between critical velocity and particle diameter is representative of nearly all classical models for particle engulfment; see, e.g., [3, 4, 5, 6, 7, 8, 9, 10]. However, the curve computed by our numerical model exhibits a slope of approximately -5/3 for particles with a diameter less than 50μm, which appears to more closely align with the experimental data. This slope becomes larger for bigger particles as the effect of gravity becomes important through sedimentation effects. Second, even though, as argued above, our model seems to reasonably describe engulfment conditions for smaller particles, we over-predict the growth velocities needed to engulf particles of diameter 45μm and larger. There are several possible reasons behind this inconsistency. For example, the presence of strong, buoyant flow in terrestrial experiments is expected to provide a lift force for larger particles that moves these particles away from the solidifying interface and delays their engulfment. In addition, of particular interest are behaviors that arise if dynamic effects are considered. We have simulated time-dependent conditions, such as oscillating thermal gradients that result in time-varying growth velocity, which are able to drive the engulfment of a particle at an average growth rate far below the critical velocity. Such heretofore unexamined dynamic mechanisms may provide an explanation for the inability of steady-state analyses to describe engulfment conditions in many systems, such as the larger particles of these

experiments. We are actively pursuing these ideas and will report our findings in a future publication.

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Publications for project

1. Y. Tao, A. Yeckel, and J.J. Derby, "Analysis of particle engulfment during the growth of crystalline silicon," J. Crystal Growth, in press (2016). doi:10.1016/j.jcrysgro.2015.12.037

2. Y. Tao, A. Yeckel, and J.J. Derby, "Steady-State and Dynamic Models for Particle Engulfment during Solidification," J. Comp. Phys., submitted (2015).

3. J. Friedrich, C. Reimann, A. Croll, T. Sorgenfrei, T. Jauss, Y. Tao, and J.J. Derby, "Interaction of SiC particles with moving solid-liquid interface during directional solidification of silicon," in preparation (2016).

4. Y. Tao, J. Friedrich, C. Reimann, A. Croll, T. Sorgenfrei, T. Jauss, and J.J. Derby, "The effect of particle lift on the engulfment of silicon carbide during the directional solidification of silicon," in preparation (2016).

Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	Tao T, Yeckel A, Derby JJ. "Analysis of particle engulfment during the growth of crystalline silicon." Journal of Crystal Growth. 2016 Oct 15;452,:1-5. (from ACCGE-20-OMVPE-17. The 20th American Conference on Crystal Growth and Epitaxy in conjunction with the 17th US Biennial Workshop on Organometallic Vapor Phase Epitaxy, August 2-7, 2015, Big Sky, MT, USA). http://dx.doi.org/10.1016/j.jcrysgro.2015.12.037 , Oct-2016
Articles in Peer-reviewed Journals	Tao T, Yeckel A, Derby JJ. "Steady-state and dynamic models for particle engulfment during solidification." Journal of Computational Physics. 2016 Jun 15;315:238-63. http://dx.doi.org/10.1016/j.jcp.2016.03.050 , Jun-2016
Articles in Peer-reviewed Journals	Tao Y, Sorgenfrei T, Jauß T, Cröll A, Reimann C, Friedrich J, Derby JJ. "Particle engulfment dynamics under oscillating crystal growth conditions." J. Cryst. Growth. 2017 Jun 15;468:24-7. https://doi.org/10.1016/j.jcrysgro.2016.10.049 , Jun-2017
Articles in Peer-reviewed Journals	Tao Y, Derby JJ. "The engulfment of a precipitated particle in a saturated melt during solidification." J. Cryst. Growth. 2022 Jan 1;577:126400. https://doi.org/10.1016/j.jcrysgro.2021.126400 , Jan-2022
Articles in Peer-reviewed Journals	Derby JJ, Tao Y, Reimann C, Friedrich J, Jauß T, Sorgenfrei T, Cröll A. "A quantitative model with new scaling for silicon carbide particle engulfment during silicon crystal growth. " J. Cryst. Growth. 2017 Apr 1;463:100-9. https://doi.org/10.1016/j.jcrysgro.2017.02.012 , Apr-2017